

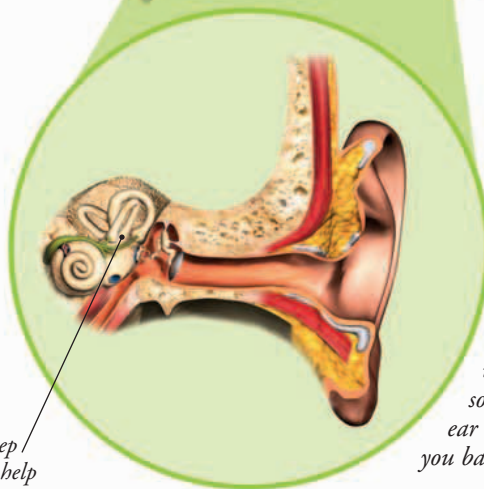
Keeping balance

Your ears allow you to hear sounds but they also give you your sense of balance. Special sensors in each inner ear keep your brain updated about how upright you are, so it can tell your body how to balance. Without your ears, you'd fall off your bike!

SIXTH SENSE

Your body has a sixth sense that keeps you on your feet automatically. The balance sensors in your ears send signals to your brain. These are supported by information from your eyes, and from position and pressure sensors in your muscles, joints, and skin. Your brain processes all this data, and sends a stream of instructions to your muscles to help you balance.

These loops deep inside the ear help monitor balance.



◀ **INSIDE YOUR EAR** Most of the organs in your ear are used for hearing, but some parts of your inner ear do the work of keeping you balanced and upright.

IN BALANCE

Inside your ears are three hollow, fluid-filled loops at right angles to each other, called semicircular canals. These are your main organs of balance. When you move your head, the fluid in these canals moves too. This movement is detected by sensory hair cells, which are wired to your brain by connecting nerves. The signals update your brain about your body's position and keep you on your feet.

